Def. Let F be a field.
Given a nonconstant polynomial $f \in F[x]$, let $\alpha \in \overline{F}$ be a root of f.
Define the multiplicity of α as a root of f is:
the largest positive integer m such that $(x-\alpha)^m$ divides f in $\overline{F}[x]$.

Lemma. Let F be a field, and let $f \in F[x]$ be an irreducible.
Then, every root of f in $\overline{F}$ has the same multiplicity.
Proof. Let $\alpha, \beta \in \overline{F}$ be roots of f. Then, there exists an isomorphism

$$\varphi : F(\alpha) \to F(\beta) \text{ such that } \varphi(\alpha) = \beta \text{ and } \varphi|_F = id_F.$$

Let L be the splitting field of f over F. Consider the extension

$$\tilde{\varphi} : L \to L \text{ such that } \tilde{\varphi}(\alpha) = \beta \text{ and } \tilde{\varphi}|_F = id_F.$$

Now, the induced map $\tilde{\varphi}_x : L[x] \to L[x]$ satisfies
$\tilde{\varphi}_x(f) = f$, because $\tilde{\varphi}$ fixes F.

Let $m \in \mathbb{N}$ be a multiplicity of α. Then,

$$f(x) = (x-\alpha)^m \cdot g \text{ for some } g \in \overline{F}[x] \text{ and } g(\alpha) \neq 0.$$

Since $\tilde{\varphi}_x$ is a ring-Homomorphism,

$$\tilde{\varphi}_x\left((x-\alpha)^m \cdot g\right) = (x-\beta)^m \cdot \tilde{\varphi}_x(g) = f$$

and $\tilde{\varphi}_x(g)(\beta) = \tilde{\varphi}_x(g)(\tilde{\varphi}(\alpha)) = \tilde{\varphi}(g(\alpha)) \neq 0$, since $g(\alpha) \neq 0$.

so that m equals to the multiplicity of β, so proved. ∎

Cor. The number of distinct roots of an irreducible polynomial $f \in F[x]$ in an algebraic closure $\overline{F}$ divides the degree of f.